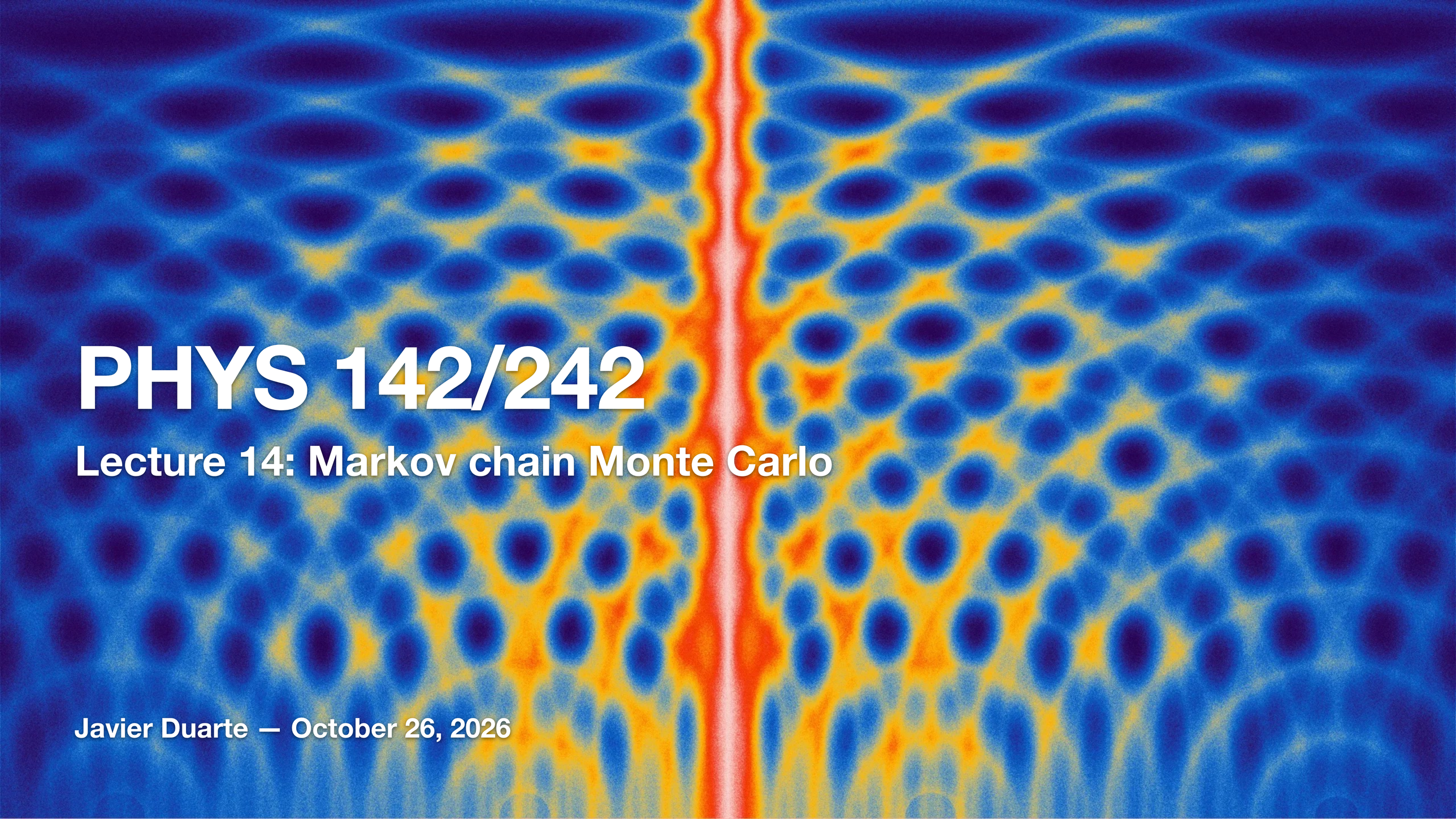


PHYS 142/242

Lecture 14: Markov chain Monte Carlo

Javier Duarte — October 26, 2026

Canonical ensemble

In the canonical ensemble, the probability p_n that the system is in a state of energy E_n is

$$p_n = \frac{1}{Z} e^{-E_n/(k_\mathrm{B}T)},$$

where $k_\mathrm{B} = 1.38 \times 10^{-23}\,\mathrm{J/K}$ is the Boltzmann constant and

$$Z = \sum_n e^{-E_n/(k_\mathrm{B}T)}$$

is the partition function.

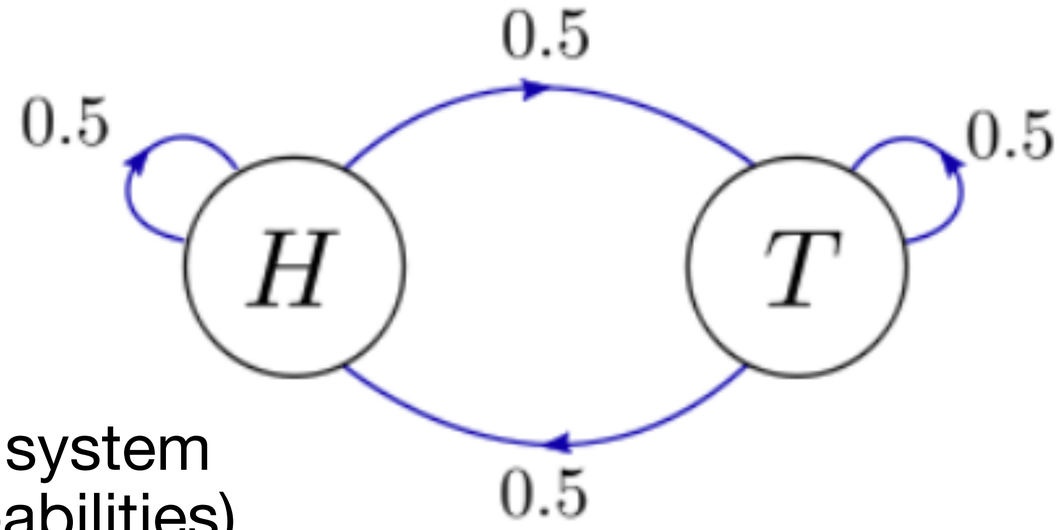
Convenient to define “inverse temperature” $\beta = 1/(k_\mathrm{B}T)$

Coin-flipping game

Suppose we have a coin which can be heads (H) or tails (T)

At each step, we flip the coin, producing a new state (H or T). This generates a sequence like "HTTHTHTTHH..."

Visualize the process with a **state diagram**



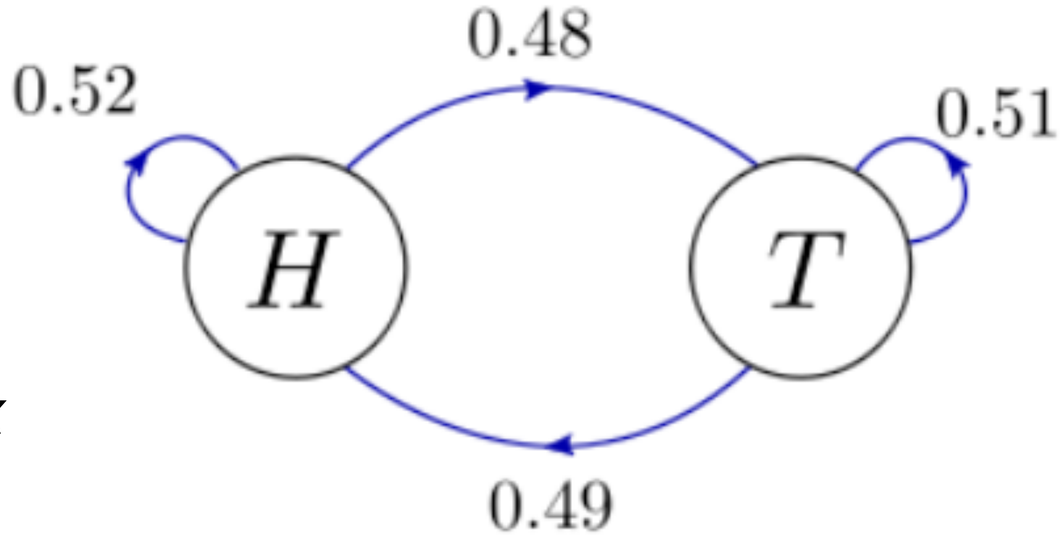
- Circles represent the possible states
- Arrows indicate the possible states that the system could transition to (with corresponding probabilities)
- Transition probabilities coming out of each state must sum up to one

Biased coin

Suppose the coin is biased to be more likely to land on the side it landed on before

State diagram looks like this. The probability of flipping a subsequent H or T is

$$p'_H = P(H|H)p_H + P(H|T)p_T$$
$$p'_T = P(T|H)p_H + P(T|T)p_T$$

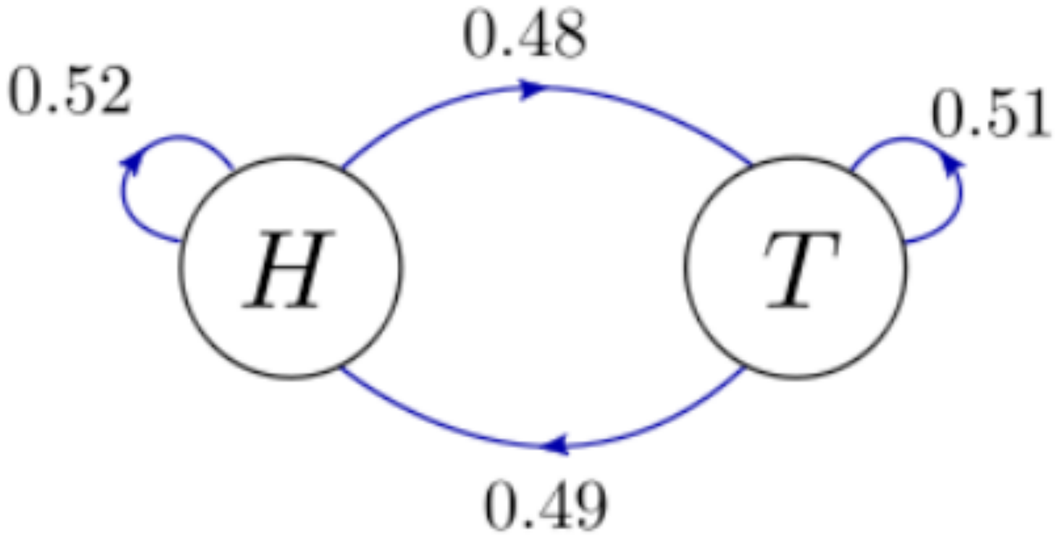


which we can write using the *transition matrix*

$$\begin{bmatrix} p'_H \\ p'_T \end{bmatrix} = \begin{bmatrix} P(H|H) & P(H|T) \\ P(T|H) & P(T|T) \end{bmatrix} \begin{bmatrix} p_H \\ p_T \end{bmatrix} = \begin{bmatrix} 0.52 & 0.49 \\ 0.48 & 0.51 \end{bmatrix} \begin{bmatrix} p_H \\ p_T \end{bmatrix}$$

Stationary state

After a large number of steps, the states probabilities might converge to a "stationary" distribution (π_H, π_T), such that they no longer change significantly on subsequent steps.



The stationary probabilities must satisfy

$$\begin{bmatrix} \pi_H \\ \pi_T \end{bmatrix} = \begin{bmatrix} P(H|H) & P(H|T) \\ P(T|H) & P(T|T) \end{bmatrix} \begin{bmatrix} \pi_H \\ \pi_T \end{bmatrix}$$

Note, this is an eigenvalue equation, with eigenvalue = 1!

Markov process and chain

A ***Markov process*** is a set of probabilistic rules that tell us how to choose a new state of the system, based only on the system's current state. If the system is currently in state n then the probability of choosing state m on the next step is given by the “transition probability” $P(m|n)$.

By repeatedly applying the Markov process, we move the system through a random sequence of states, $\{n^{(0)}, n^{(1)}, n^{(2)}, \ldots\}$ where $n^{(k)}$ denotes the state on step k.

This random sequence is called a ***Markov chain***.

Because the system must transition to some state,

$$\sum_m P(m|n) = 1 \quad \text{for all } n \in \{0,1,\ldots\}\,.$$

Markov process and chain

The probability to be found in state m at step $k+1$ depends only on step k

$$p_m^{(k+1)} = \sum_n P(m\,|\,n)\, p_n^{(k)}$$

or in terms of the transition matrix,

$$\begin{bmatrix} p_0^{(k+1)} \\ p_1^{(k+1)} \\ \vdots \end{bmatrix} = \begin{bmatrix} P(0\,|\,0) & P(0\,|\,1) & \cdots \\ P(1\,|\,0) & P(1\,|\,1) & \cdots \\ \vdots & \vdots & \end{bmatrix} \begin{bmatrix} p_0^{(k)} \\ p_1^{(k)} \\ \vdots \end{bmatrix},$$

The vector of stationary probabilities $[\pi_0, \pi_1, \pi_2, \ldots]$ is an eigenvector of the transition matrix, with eigenvalue 1.

We will use Markov processes to model thermodynamic systems, such that a stationary distribution represents the distribution of thermodynamic microstates in thermal equilibrium

Detailed balance

One way to figure out π_n is through the property of detailed balance.

Detailed balance requires that the rate at which the $n \to n+1$ transition occurs is the same as the rate at which the opposite transition occurs $n+1 \to n$.

$$P(n+1\,|\,n)\,\pi_n = P(n\,|\,n+1)\,\pi_{n+1} \quad \text{for all} \quad n \in \{0,\ldots,N\}\,.$$

If this holds for every pair of states, then the probability distribution is necessarily stationary.

$$\begin{bmatrix} \pi_0 \\ \pi_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} P(0\,|\,0) & P(0\,|\,1) & \cdots \\ P(1\,|\,0) & P(1\,|\,1) & \cdots \\ \vdots & \vdots & \end{bmatrix} \begin{bmatrix} \pi_0 \\ \pi_1 \\ \vdots \end{bmatrix},$$

Note: not all Markov chains obey detailed balance!

Example: 2D Ising model

Set of lattice sites Λ, usually a square grid

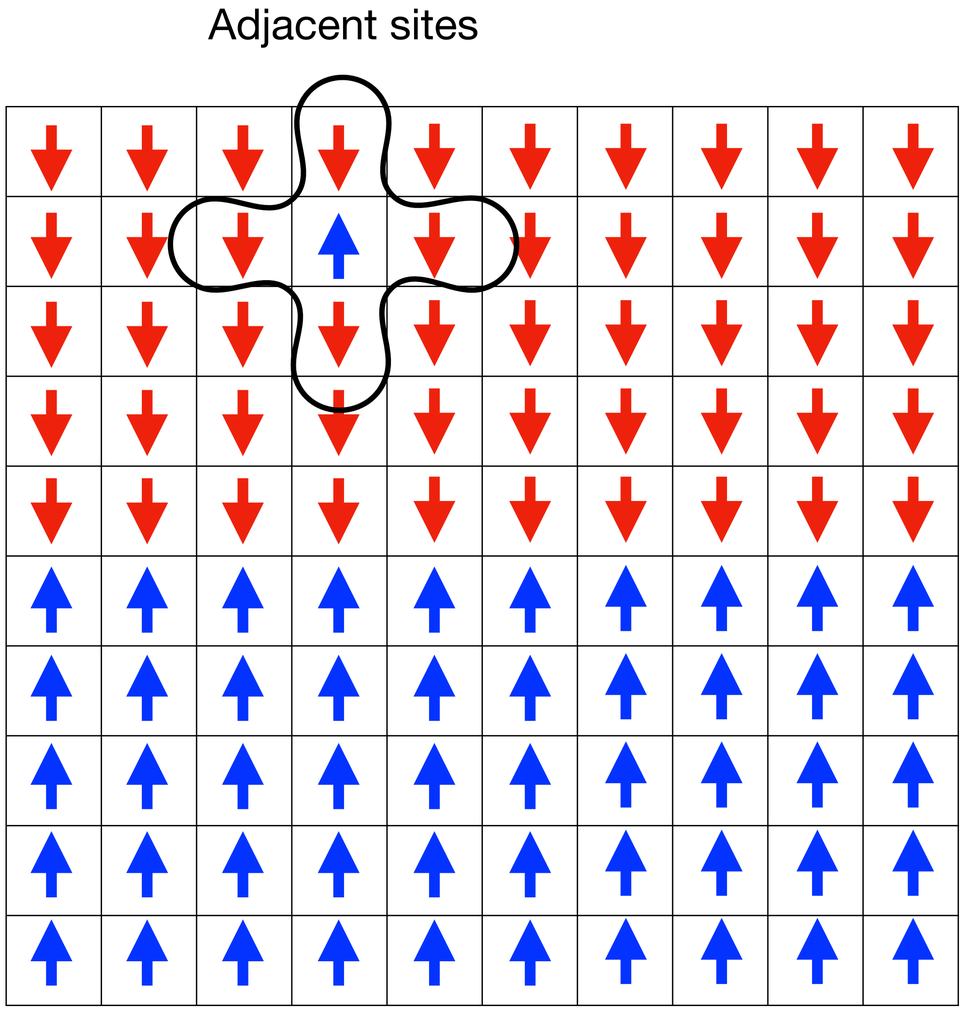
At each site, there is a discrete variable $\sigma_k \in \{-1, +1\}$, representing the site's spin

A spin configuration $\sigma = \{\sigma_k\}_{k \in \Lambda}$

For any two adjacent sites $\langle ij \rangle$ (no double counting) there is a spin interaction J_{ij} and at every site there may be an external field h_j

$$E(\sigma) = -\sum_{\langle ij \rangle} J_{ij}\sigma_i\sigma_j - \mu \sum_j h_j\sigma_j$$

Simplification: $J = J_{ij} > 0$ and $h_j = 0$



Energy

Given the simplified Hamiltonian

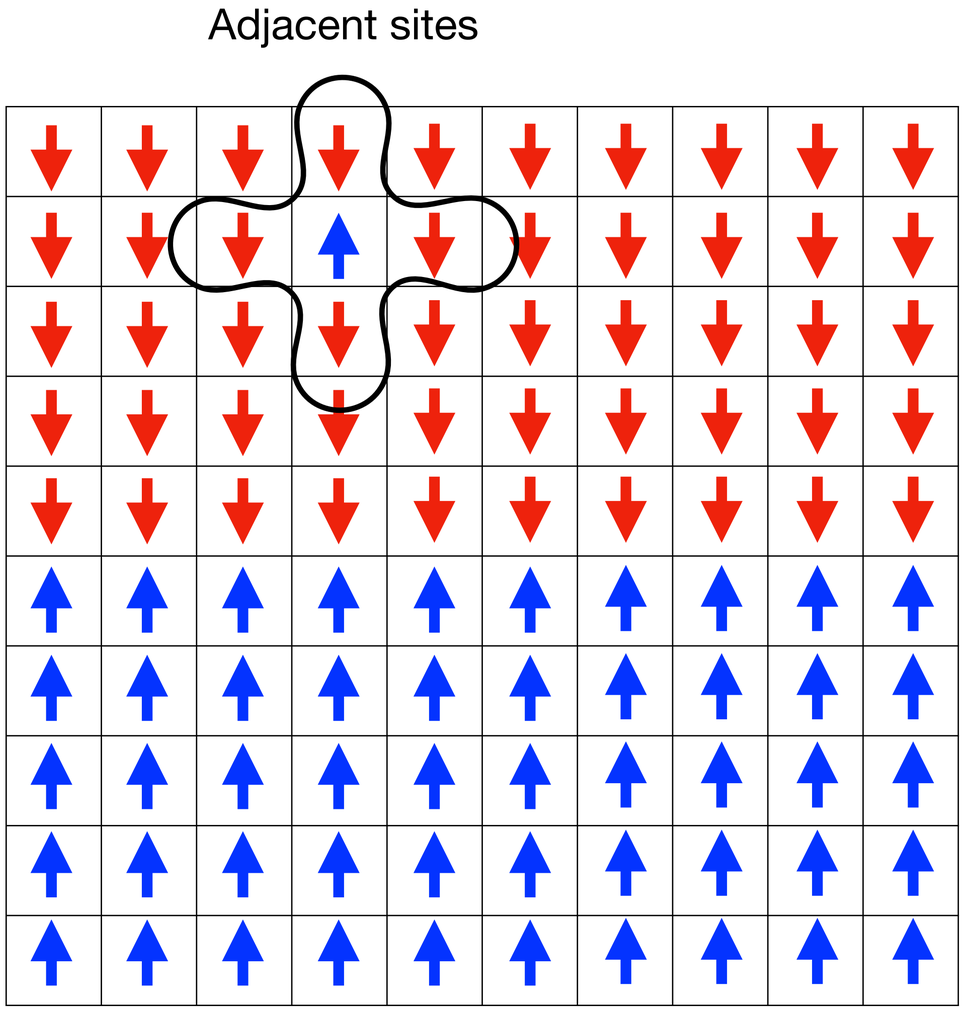
$$E(\sigma) = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

The energy of the configuration here is
$E = -152J$

Assume periodic boundary conditions

Hint: number of adjacent site pairs (no double counting) is

$$N_{\text{adjacent pairs}} = 2|\Lambda|$$



Probability

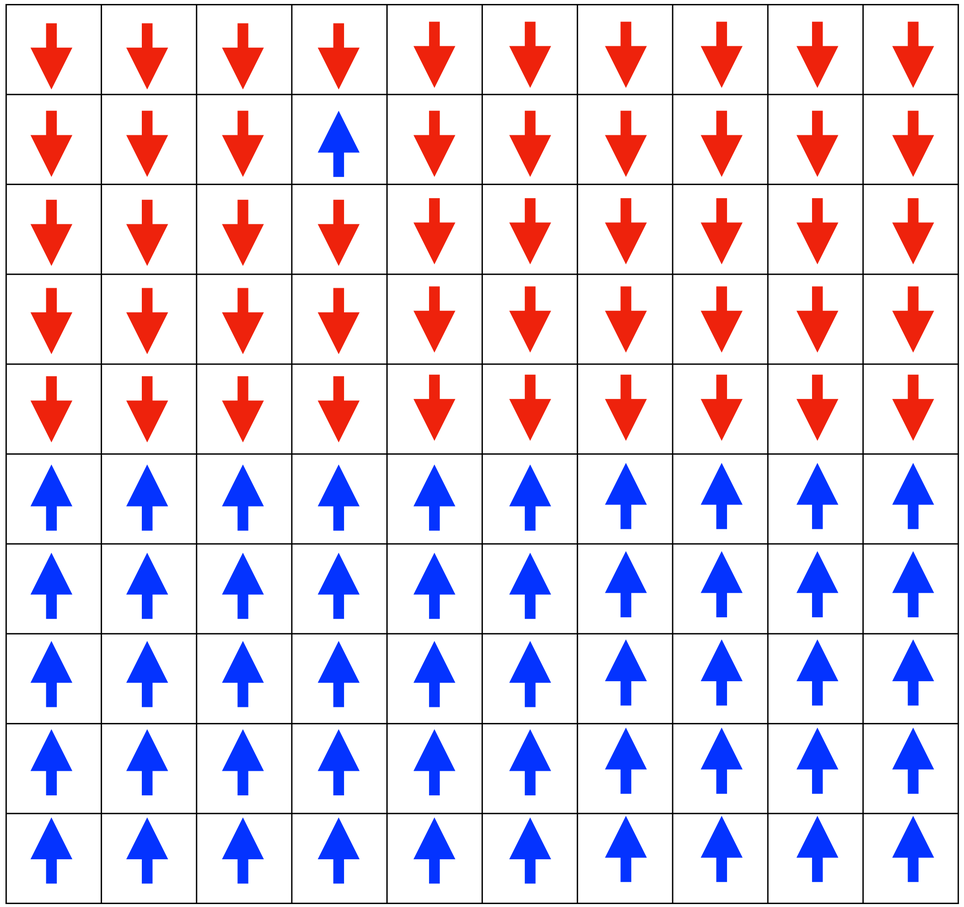
The probability of any given configuration follows the Boltzmann distribution

$$p(\sigma) = \frac{1}{Z} \exp\left(-\frac{E(\sigma)}{k_\mathrm{B}T}\right)$$

where $Z = \sum_\sigma \exp\left(-\frac{E(\sigma)}{k_\mathrm{B}T}\right)$

is the partition function.

With 10×10 sites $2^{100} \approx 10^{30}$ possible states

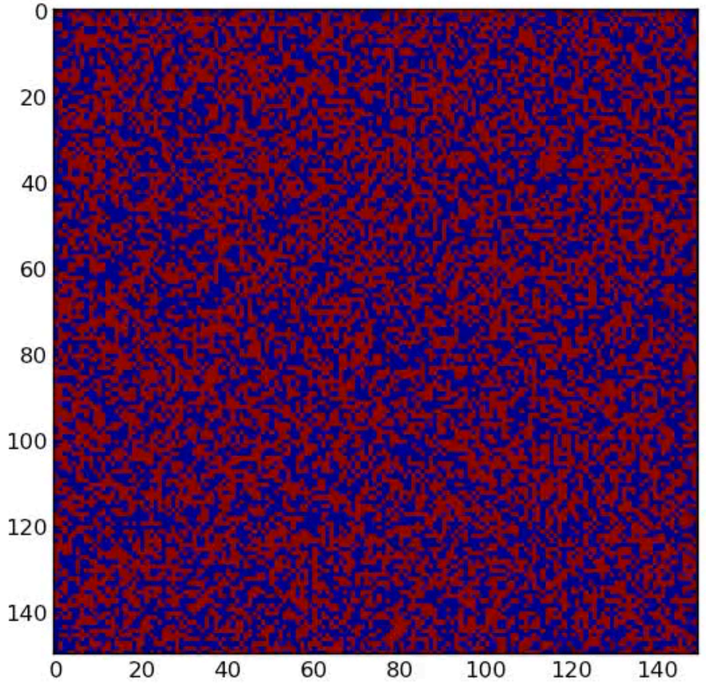


Ising model phase transition

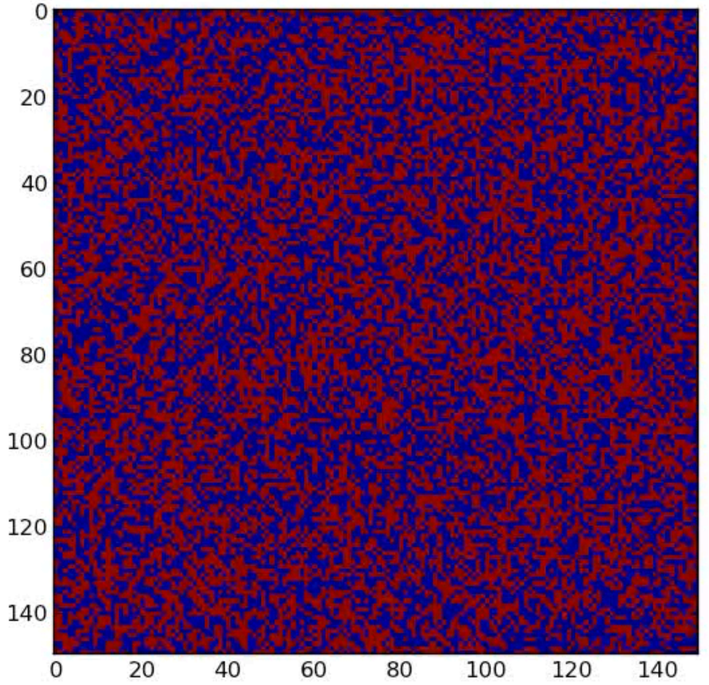
With 150×150 lattice, $2^{22500} \approx 10^{6773}$ possible states!

Phase transition occurs at a critical temperature: ordered phase at low T, disordered phase at high T. How can we compute this?

High T



Low T



Ising model phase transition

Often interested in quantities like the average magnetization, i.e. first we average over the lattice

$$\sigma_{\text{ave}} = \frac{1}{|\Lambda|} \sum_{k \in \Lambda} \sigma_k$$

then we take a thermodynamic average

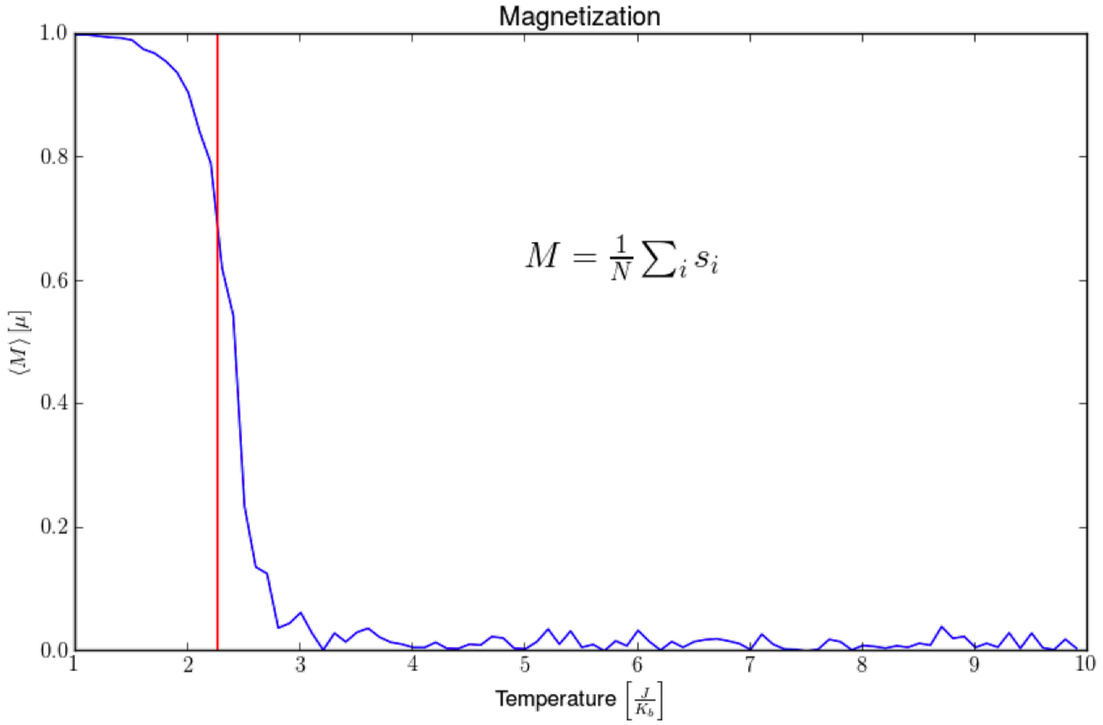
$$\langle \sigma_{\text{ave}} \rangle = \sum_{\sigma} \sigma_{\text{ave}} p(\sigma)$$

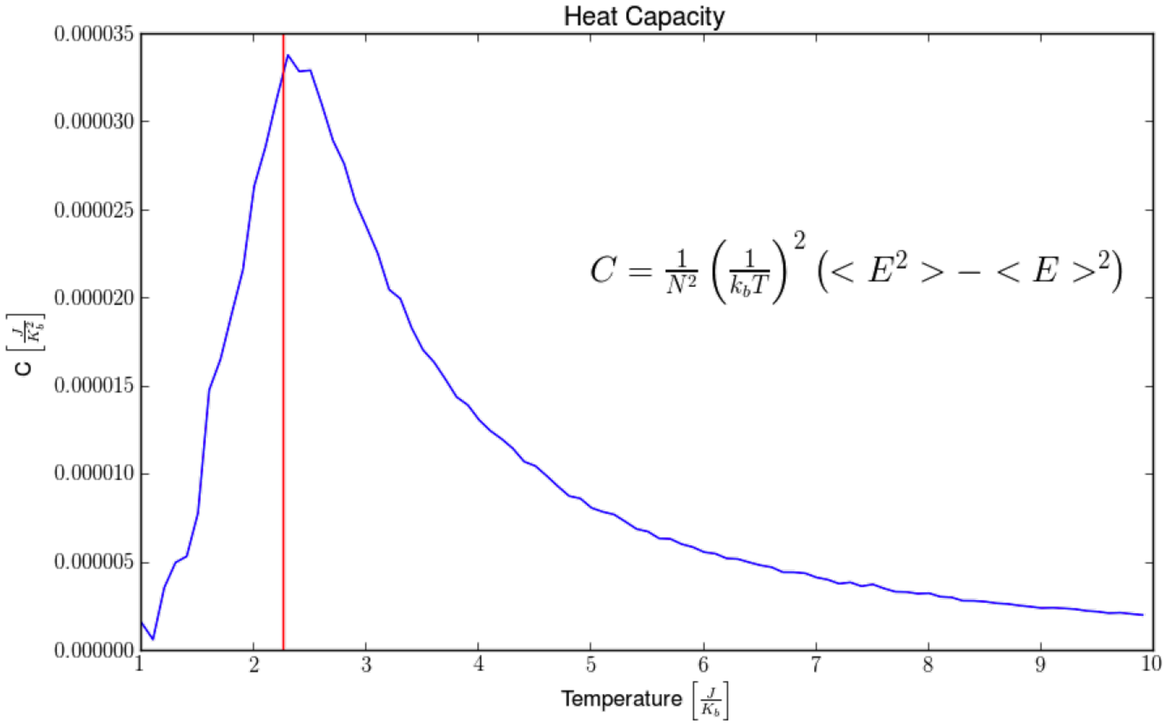
If we plot this versus temperature, we can see there is some critical temperature $T_{\text{c}}$where the system goes from being ferromagnetic (spins are aligned) to paramagnetic (spins are not aligned)

Ising model phase transition

In 2D, we can actually analytically calculate the critical temperature as solved in L. Onsager, Phys. Rev. 65, 117 (1944)

$$\frac{k_{\mathrm{B}}T_{\mathrm{c}}}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$$





Markov chain MC

Trying to calculate the partition function directly by summing over all states is prohibitively time-consuming.

Instead the MCMC method selectively samples the states.

To accomplish this, we design a Markov process whose stationary distribution is equal to the correct distribution of probabilities.

Once we have this Markov process, we can generate a long Markov chain and calculate moving averages of our desired quantities like $\langle \sigma_{\text{ave}} \rangle$.

If the Markov chain is long enough, this will converge to the true expectation value

Metropolis algorithm

Design a Markov process to match a given stationary distribution π_n: ***Metropolis algorithm*** based on detailed balance

$$P(n\,|\,m)\,\pi_m = P(m\,|\,n)\,\pi_n \quad \text{for all } m, n$$

1. On step k, the system is in state n. Randomly choose a candidate state m by making an unbiased random step through the space of possible states.

2. Compare the probabilities π_n and π_m

   If $\pi_m \geq \pi_n$, accept the candidate

   If $\pi_m < \pi_n$, accept the candidate with probability π_m/π_n. Otherwise, reject the candidate

3. If the candidate is accepted, the state in step $k+1$ is$m$. Otherwise, the state in step $k+1$ remains n

4. Repeat

Checking detailed balance

Let's verify that the stationary distribution of the Markov process satisfies detailed balance.

Consider two states a, b with $\pi_a \leq \pi_b$.

Starting from a, suppose we choose candidate step $a \to b$ with some probability q. Then, the transition probability for $a \to b$ is q times the acceptance probability 1.

Starting from b, we will choose candidate step $b \to a$ with the same probability q (assuming this is ***unbiased***). Then, the transition probability for $b \to a$ is q times the acceptance probability π_a / π_b.

$$\left.\begin{aligned} P(b \,|\, a) &= q \\ P(a \,|\, b) &= q \frac{\pi_a}{\pi_b} \end{aligned}\right\} \quad \Rightarrow \quad P(a \,|\, b)\, \pi_b = P(b \,|\, a)\, \pi_a \quad \text{(detailed balance)}$$

The Hastings correction

Slide 17 assumed an ***unbiased*** proposal, $q(m|n) = q(n|m)$. If the proposal is biased, Metropolis samples the wrong distribution. The fix is to accept the candidate with probability

$$A(m|n) = \min\left(1,\ \frac{\pi_m\, q(n|m)}{\pi_n\, q(m|n)}\right)$$

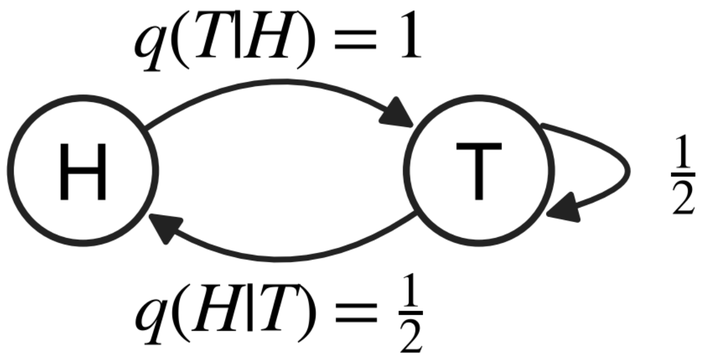
This satisfies detailed balance for any proposal:

$$P(m|n)\,\pi_n = q(m|n)\,A(m|n)\,\pi_n = \min\left(q(m|n)\,\pi_n,\ \ q(n|m)\,\pi_m\right)$$

The right side is unchanged when $m \leftrightarrow n$, so $P(m|n)\ \pi_n = P(n|m)\ \pi_m$. With an unbiased proposal the q's cancel, and this is the Metropolis rule from slide 16.

A biased proposal for the coin

Target: (π_H, π_T) = (⅓, ⅔)



From T, the proposal picks H half the time and stays at T otherwise.

	Metropolis	Hastings
P(T|H)	1	1
P(H|T) = q(H|T) A(H|T)	½ · ½ = ¼	½ · 1 = ½
Flow H → T: π_H P(T|H)	⅓	⅓
Flow T → H: π_T P(H|T)	⅙, unbalanced	⅓, balanced
Chain samples	(⅕, ⅘): wrong	(⅓, ⅔): correct

Metropolis accepts T → H with min(1, π_H/π_T) = ½. Hastings multiplies that ratio by q(T|H)/q(H|T) = 2, so it accepts every proposed move.

Stepping through state space

One way of thinking about the Metropolis algorithm is that it takes a scheme for performing an unbiased random walk through the space of possible states (represented by our candidate choices), and converts it into a scheme for performing a biased random walk

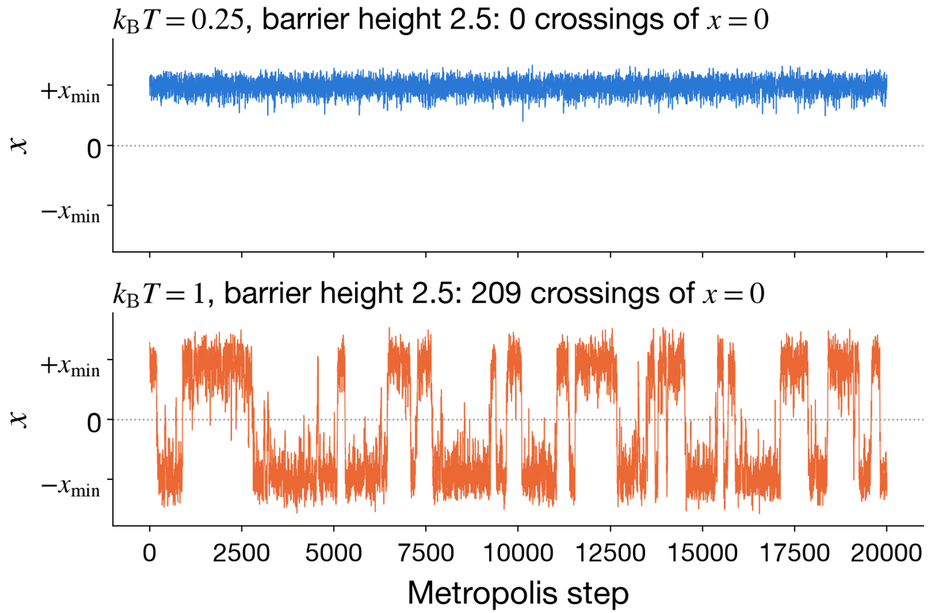
The biased random walk corresponds to a Markov process with the stationary distribution we are interested in

Detailed balance is not enough

Irreducible

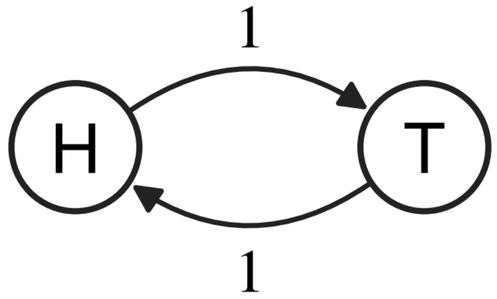
Every state must be reachable from every other. Otherwise time averages converge to the wrong answer.

In practice: a double well at low T can cross the barrier in principle, but almost never does.



Aperiodic

The chain must not cycle. Otherwise the distribution at step k never settles, and burn-in loses its meaning.



This coin always flips. π = (½, ½) satisfies detailed balance, yet a chain started at H alternates H, T, H, T, … forever.

Rejections, where the chain stays put, keep Metropolis aperiodic.

Applying MCMC to the 2D Ising model

To apply the MCMC method to the Ising model, we design a Markov process using the Metropolis algorithm as follows

1. On step k, randomly choose one of the spins i and consider flipping it $\sigma_i \rightarrow -\sigma_i$
2. Calculate the change in energy that would result from flipping spin i, i.e. the quantity:

$$\Delta E = -\left[J \sum_{\langle ij \rangle} \sigma_j \right] \Delta\sigma_i,$$

where $\Delta\sigma_i = -2\sigma_i$ is the change in σ_i due to the spin flip

If $\Delta E \leq 0$, accept the spin flip

If $\Delta E > 0$, accept the spin flip with probability $\exp(-\Delta E/k_{\mathrm{B}}T)$. Otherwise, reject the flip.

3. Update the moving average of $\langle \sigma_{\mathrm{ave}} \rangle$ (or whatever quantity we are interested in).

4. Repeat.

Practical considerations for MCMC

- Several practical considerations
  - Acceptance rate: the rate at which we accept proposals is important. We want this to be relatively high (to reduce total running time), but not always 1 (which may mean we're not sufficiently sampling the full space). Typical values are 15% to 50%
  - Burn in
  - Error and (integrated) autocorrelation time

Burn-in

"Burn-in": common to throw out the first few states of a Markov chain, maybe the first 100 or the first 1000

The idea is to get rid of "transient behavior" connected to an improbable intial configuration

Equivalent: pick an initial state near equilibrium (i.e. not unlikely)

Autoregressive series

$$x_{n+1} = rx_n + e_n$$

with $r = 0.98$ and

$$e_n \sim \mathcal{N}(0,1)$$

starting from $x_0 = 10$ (left)

and $x_0 = 0$ (right)

